

Introduction to Machine Learning

& Data Science Workflows

Applied Machine Learning — Session 1, Chapter 1

What Is Machine Learning?

Traditional programming Rules + Data $\rightarrow$ Output

Machine Learning Data + Output $\rightarrow$ Rules

"A computer program is said to learn from experience E with respect to some task T and performance measure $P...$ " – Tom Mitchell, 1997

Why Machine Learning?

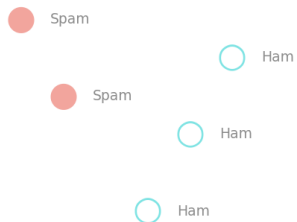
- Email spam → millions of patterns, impossible to hand-code
- Image recognition → pixels to meaning
- Medical diagnosis → beyond human attention span
- Recommendations → individual preferences at scale

Key insight: When rules are too complex to write — let data write them.

Three Paradigms of ML

Three Paradigms of Machine Learning

Supervised Learning (X, y) → predict y



Labels provided → Model learns boundary

Unsupervised Learning X only → find structure



No labels → Algorithm discovers groups

Reinforcement Learning trial & error → max reward



Goal: maximize cumulative reward

Supervised Learning

- Training data has **labels**
- Model learns: $X \rightarrow y$
- Two flavors:
 - **Regression** $\rightarrow$ continuous output (price, temperature)
 - **Classification** $\rightarrow$ discrete output (spam/not spam, disease/no disease)

Unsupervised Learning

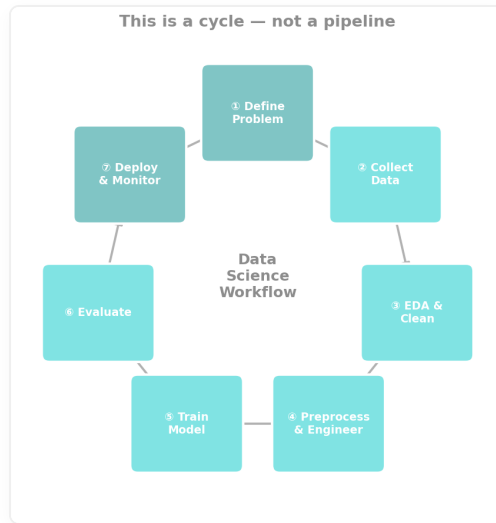
- **No labels** — only raw data
- Model finds hidden structure
- Applications:
 - Customer segmentation
 - Anomaly detection
 - Dimensionality reduction
 - Topic modeling

Reinforcement Learning

- **Agent** interacts with **Environment**
- Takes **Actions** → receives **Rewards**
- Goal: maximize cumulative reward

Agent $\rightarrow$ Action $\rightarrow$ Environment
 $\uparrow$ $\downarrow$
 $\leftarrow\leftarrow\leftarrow$ Reward + State $\leftarrow\leftarrow\leftarrow$

The Data Science Workflow



This is a cycle, not a pipeline.

Step ① — Define the Problem

- What are we trying to predict/discover?
- Who uses the result?
- What is a "good enough" answer?

Most ML project failures happen here, not in the model.

Step ② — Data

- Where does it come from?
- Is it representative?
- How much do we have?
- What are the legal/ethical constraints?

Garbage in → Garbage out.

Steps ③④ — EDA & Preprocessing

- Explore distributions, relationships, outliers
- Handle missing values
- Encode categorical features
- Scale numerical features
- Split: train / validation / test

~80% of project time lives here.

Steps ⑤⑥ — Train & Evaluate

- Choose algorithm(s)
- Fit on training data
- Measure performance on held-out data
- Avoid overfitting

Step ⑦ — Deploy & Iterate

- A model in a notebook helps no one
- Deployment = value creation
- Monitor: data drifts, model degrades
- Retrain, improve, repeat

Why Iteration Matters

Most projects fail not because of bad algorithms — but because of bad iterations.

What goes wrong	How iteration fixes it
Wrong problem definition	Early feedback from stakeholders
Dirty / biased data	EDA catches this before training
Model doesn't generalize	Evaluation reveals overfitting
Features miss signal	Domain experts improve features
Production drift	Monitoring triggers retraining

The cycle is the method. Expect at least 3 full loops.

The Python ML Ecosystem

Tool	Role
<code>numpy</code>	Arrays & math
<code>pandas</code>	Data wrangling
<code>matplotlib</code> / <code>seaborn</code>	Visualization
<code>scikit-learn</code>	Algorithms & pipelines
<code>gymnasium</code>	RL environments

Now: Examples!

→ Open `ch01_introduction_examples.ipynb`

We will:

1. Load the famous **Iris dataset**
2. Inspect its structure
3. Visualize features
4. Build first intuitions — **before any model**

Key Takeaways

- ML = learning patterns from data automatically
- Three paradigms: supervised / unsupervised / reinforcement
- Workflow is a **cycle** — expect iteration
- Great ML starts with **understanding the data**

Next: Chapter 2

Data Selection, Cleaning & Preparing

"Real-world data is messy, incomplete, and full of surprises. Let's learn how to tame it."